

LOUNES BENALI

DEEP LEARNING ENGINEER | SPEECH & MULTIMODAL AI | LARGE-SCALE TRAINING & INFERENCE



B-Lounes

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[Portfolio](#)

PROFILE

Speech and multimodal AI engineer working across large-scale data curation, distributed training, rigorous evaluation, and low-latency inference. Currently co-developing a multilingual ASR model centered on Arabic dialects at MBZUAI's Institute of Foundation Models on Slurm-managed AMD MI210 and NVIDIA H200 clusters.

PROFESSIONAL EXPERIENCE

Speech AI Research Intern

Feb 2026–Aug 2026 (expected)

Institute of Foundation Models (IFM), MBZUAI

Paris, France

Open foundation-model research lab behind K2, K2 Think, and the PAN world model.

- Core contributor in a four-person speech team developing a multilingual ASR model. Its first checkpoint achieves the best CER and third-best WER on the public Open Universal ASR Leaderboard, within 2 WER points of the leaders (Cohere and Audar) while also supporting diacritized transcription, dialect identification, and code-switching with French, English, Spanish, and other languages; emotion-tag support is under development.
- Developed a large-scale audio-scraping pipeline that collected nearly 4M hours, with deterministic VPN-configuration management, RAM-backed intermediate I/O to avoid disk bottlenecks, and worker scheduling that kept CPU resources saturated.
- Built a multi-model speech-data curation pipeline over 220M segments (approximately 1M hours), combining dialect-ID, quality labels, ASR hypotheses, and token confidence to distill a stringently filtered 17k-hour ASR training corpus.
- Built and operated Slurm workflows across AMD MI210/ROCm and NVIDIA H200 clusters; scaled distributed training to 256 GPUs and launched fault-tolerant inference on 1,440 MI210 GPUs, with checkpoint/resume, health monitoring, and automatic recovery.
- Ported speech training and inference stacks to ROCm, building a native vLLM environment, patching multimodal audio execution paths, and resolving k2 GPU ragged-label failures for reliable training and serving on MI210 GPUs.
- Ran from-scratch self-supervised VietASR pretraining for ASR, using iterative k-means pseudo-labeling and fine-tuning with as little as 100 hours of labeled speech; scaled runs across 8–16 nodes and evaluated on a balanced 19-dialect set.
- Reimplemented the production Kaldi/FSMN VAD path as batched GPU operations, reaching $9,668\times$ real time on 8 MI210 GPUs while reproducing all 32,794 CPU-reference boundaries within 10 ms.

Deep Learning Engineer (Apprentice)

Sep 2024–Feb 2026

Ampere Software Technology (Renault Group)

Guyancourt, France

- Developed and fine-tuned text-to-speech (TTS) and speech-to-speech systems within an applied R&D team.
- Built an internal inference platform serving multiple fine-tuned TTS models, voices, versions, and languages to several company teams; implemented real-time streaming, model compression, inference acceleration, and latency optimization.
- Contributed across the full deep-learning lifecycle, including dataset generation and collection, data processing, fine-tuning implementation, and production deployment.

Technical Staff – Paris 2024 Olympic Games

May 2024–Sep 2024

International Broadcast Center (IBC)

Le Bourget, France

- Deployed and configured computers, monitors, printers, and other networked devices on the IBC LAN.
- Diagnosed and resolved network and endpoint issues to support reliable daily broadcast operations.

Engineering Intern

Jul 2023

Schneider Electric

Cheraga, Algeria

- Studied variable-frequency-drive architectures and explored opportunities for machine-learning-based control optimization.

SELECTED PROJECTS

Conversational AI & Synthetic Speech Dataset Generation Python, TypeScript, real-time audio

- Developed a modular platform orchestrating multi-agent, real-time voice conversations to generate aligned synthetic speech datasets.
- Engineered cross-agent audio routing with speaker-separated 16-bit PCM streams at 24 kHz, plus resilient re-tries and connection recovery; exported aligned transcripts, per-speaker channels, turn-taking annotations, and metadata for TTS and speech-to-speech post-training.

Seenitt – Streaming Visual AI for Smart Glasses Python, TensorRT, CUDA, OpenCV, [GitHub](#)

- Received a MentraOS developer grant to develop Seenitt, a streaming visual-AI system for smart glasses; built a Python WebSocket backend that decodes live H.264 camera feeds and returns structured results to the client.
- Implemented a real-time chess assistant using SAM 3 to segment a physical chessboard and its pieces, reconstruct and track its 9×9 grid, stabilize observations against legal moves, and return FEN updates with Stockfish suggestions.
- Profiled the FP16 TensorRT/PyCUDA path, cached both text prompts, and reused each frame’s vision-encoder output for board and piece segmentation; tuned CUDA kernels and memory transfers for per-frame latency.
- Ran geometric grid reconstruction while the GPU decoded piece masks; tracked the 81 grid intersections between frames with Lucas–Kanade optical flow and triggered a full reconstruction only when tracking failed.
- Prototypes under development: a plant-species identification assistant combining prompt-driven segmentation with ViT classification and best-view cataloguing; and a Blackjack assistant that detects cards, groups player hands, and provides rule-based action recommendations.

Computer Vision & 6-DoF Vehicle Pose Estimation PyTorch, CenterNet, OpenCV, MATLAB

- Developed an hourglass CenterNet pipeline for 6-DoF vehicle pose estimation, using perspective, camera-rotation, and color augmentations to improve robustness.
- Implemented camera calibration, homography and pose estimation, visual tracking, test-time augmentation, model blending, and quantitative error analysis.

EDUCATION

Master’s Degree in Intelligent Systems Engineering Sep 2024–Present
Sorbonne University *Paris, France*

Bachelor’s Degree in Electrical Engineering and Computer Science Sep 2023–May 2024
Paris-Saclay University *Paris, France*

- Graduated top of the class.

Engineering Cycle – Electronics Engineering and Computer Science Sep 2022–Jun 2023
National Polytechnic School of Algiers *Algiers, Algeria*

Preparatory Classes in Science and Technology Sep 2020–Jun 2022
Two-year intensive national competitive-exam program *Algeria*

- Ranked 29th out of 1,576 candidates in the national competitive entrance examination for engineering schools.

TECHNICAL SKILLS

Speech & Multimodal AI: ASR, TTS, speech-to-speech, speech SSL, PyTorch, k2, Hugging Face, WER/CER
Distributed ML & Inference: Slurm, DDP, FSDP, vLLM, AMD ROCm/MI210, NVIDIA H200/CUDA, multi-node training and inference

Vision & Edge Inference: SAM 3, TensorRT, ONNX Runtime, OpenCV, YOLO, quantization

Programming: Python, C, C++, TypeScript, Kotlin, MATLAB, VHDL

Data & Systems: PostgreSQL, DuckDB, Parquet, WebDataset, Docker, Git, REST APIs, WebSockets

Hardware & Embedded: FPGA, Raspberry Pi, PCB design

LANGUAGES & ACTIVITIES

Languages: French (DALF C2); English, Berber, and Arabic (fluent)

Activities: IT member, IEEE Student Branch; Micromouse team lead, National Polytechnic School of Algiers